Subject Code	AF6208
Subject Title	Econometrics Methods
Credit Value	3
Level	6
Normal Duration	One Semester
Pre-requisite / Co- requisite/ Exclusion	None
Objectives	This subject contributes to the achievement of the DBA outcome by sharpening students' ability to conduct original applied research and ethical awareness in business administration, laying the foundation for lifelong learning within Hong Kong and global contexts (Outcome 3).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the basic concepts, the linear regression models used to conduct empirical research in economics, finance and accounting. b. Understand some advanced econometric theories and models which can be used for advanced empirical research. c. Understand how to transform ideas and hypotheses of a DBA thesis into testable econometric hypotheses.
Subject Synopsis/ Indicative Syllabus	Econometric methods widely used in empirical research are included. • Multiple linear regression models: least squares estimation, omitted variable problem, dummy variables, heteroscedasticity. • Endogeneity problems: Instrumental variables, two stage least squares estimation, Hausman test. • Qualitative and limited dependent variable models: logit/ probit models, self-selection models, Tobit model. • Panel data models: fixed vs random effects models. • Methods for time series data: autocorrelation, ARCH, unit roots, cointegration, vector autoregressive (VAR) models
Teaching/Learning Methodology	This will be based on a series of lectures and seminars. Focusing on the practical use of quantitative research methods, the lectures will introduce econometric models with a minimum level of theoretical discussion. The lectures will also explain how to use computer software to estimate the models and how to interpret and report computer results. Participants are required to conduct empirical research by applying the research methods to contemporary issues.

Assessment Methods in Alignment with Intended Learning Outcomes	

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
		a	b	c
Continuous assessment*	**100%**	✓	✓	✓
1. Class participation	20%	✓	✓	✓
2. Individual/group research report or homework assignments	30%	✓	✓	✓
3. Take-home examination	50%	✓	✓	✓
Total	**100 %**			

**Weighting of assessment methods/tasks in continuous assessment may be different, subject to each subject lecturer*

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Class participation and interaction are a necessary means of assessment as it will provide good feedback to each individual classmate on his/her research and consultancy ideas. The experience sharing session in the workshop will be assessed by class participation. It will help clarify the concepts, methodology and critical success factors in performing research work and consultancy project.

Individual/group research report is designed to train students to learn how to conduct practical research work on his/her own. Each student will take initiative to discuss research ideas with classmates and lecturers, and decide on the design of a specific research topics suitable for further exploration. Each student is required to write a report on his/her research plan. By such assessment, it is expected that his/her understanding on the concepts of quantitative approaches to research will be enhanced.
Homework assignments are designed to help students understand the concepts and techniques of econometrics for their use in their future research.

The take-home exam is designed to check the students' understanding of the research methods as well as the ability of using computer software.

Student Study Effort Expected		
	Class contact:	
	▪ Lectures	30 Hrs.
	Other student study effort:	
	▪ Preparation for lectures and assignment	20 Hrs.
	▪ Review of lectures	60 Hrs.
	Total student study effort	110 Hrs.

Reading List and References

Textbooks

Wooldridge, J.M., Introductory Econometrics: A Modern Approach (Latest edition), South-Western

Greene, W.H., Econometric Analysis (latest edition), Pearson

Published Papers (selected)

Adams, R.B., and D. Ferreira, 2009, Women in the boardroom and their impact on governance and performance, *Journal of Financial Economics*, 94, 291–309

Bushee, B. J., 1998, The influence of institutional investors on myopic R&D investment behavior, *Accounting Review*, 73, 305-333

Campbell, John Y., Martin L., Malkiel B.G. and Xu, Y., 2001, Have individual stocks become more volatility? An empirical exploration of idiosyncratic risk, *Journal of Finance,* 56, 1 – 43

Hausman J.A., 1978, Specification tests in econometrics, *Econometrica*, 46, 1251-1271

Konchitchki Y., and P. N. Patatoukas, 2014, Accounting earnings and gross domestic product, *Journal of Accounting and Economics*,57, 76 – 88

Larcker, D. F., and T. O. Rusticus, 2010, On the use of instrumental variables in accounting research, *Journal of Accounting and Economics,* 49, 186–205

Li, D., Nguyen, Q. N., Pham, P.K. and Steven X. Wei, 2011, Large foreign ownership and firm-level stock return volatility in emerging markets, *Journal of Financial and Quantitative Analysis*, 46, 1127-1155

Newey, W.K., and K. D., West, 1987, A simple positive semi-definite heteroskedasticity and autocorrelation consistent covariance matrix, Econometrica, 55, No. 3, 703-708

Petersen, M.A., 2009, Estimating standard errors in finance panel data sets: Comparing approaches, *Review of Financial Studies*, 22, 435 – 480

Tobias, B, B. Valentin, G. Ana and P., Manju (2020). On the rise of FinTech: Credit Scoring Using Digital Footprints, *Review of Financial Studies* 33, 2845-2897.

Wei, Steven X., and Zhang C., 2006, Why did individual stocks become more volatile? Journal of Business 259 - 294

Computer Software

SAS (recommended), STATA or SPSS